

REST Endpoints & RabbitMQ Topology

Core Actor Roles & System Responsibilities

1. **System Administrator:** Manages technical platform operations, system configuration parameters, active maintenance banners, notification templates, data backups/restorations, staff 2FA invitations, and monitors the immutable, IP/timestamp-logged audit trail via the Audit & Storage Service. Can only register Admin and Management accounts via invitation link.
2. **Faculty Management:** Oversees strategic platform governance and quality assurance across both Vacancy Portal and Event Management. Reviews and approves/rejects Industry Partner account registrations and job vacancy submissions. Audits per-user change logs, generates faculty-level accreditation/QA reports, and monitors executive analytics for placement rates and industry engagement.
3. **Faculty Coordinator:** Owns Event Management lifecycles end-to-end (creation, scheduling, venue allocation, cancellation, certificate eligibility rules). Maintains deduplicated directories of guest speakers and partner companies. Reviews student event registration queues to enforce capacity limits, monitors live dynamic QR check-ins, reviews post-session feedback, and triggers certificate issuance.
4. **Academic Staff:** Supports student event participation by browsing upcoming sessions, registering students on their behalf, verifying attendance, reviewing student participation histories and feedback, and pulling departmental attendance reports.
5. **Administrative Staff:** Handles operational event logistics (venue bookings, participant list exports, speaker contact maintenance), performs manual QR check-in overrides for edge cases, sends event notifications/reminders, and physically prepares/distributes certificates once eligibility is confirmed.
6. **Guest Speaker:** Authenticated actor receiving a secure single-use email invitation (accept/decline). Accesses assigned session details (date, time, venue, agenda, expected audience size), updates personal profile, uploads presentation slides prior to events, receives automated 2-day pre-session reminders (with live registration counts), and views anonymized post-session participant feedback summaries.
7. **Student (Undergraduate):** Auto-provisioned via university enrollment database synchronization (POST /api/v1/auth/students/sync). Uses a single authentication session across all modules to browse events, register (subject to capacity approval), perform live QR check-ins, submit feedback, claim PDF certificates, upload multiple resumes, toggle job search availability (ACTIVELY_LOOKING / NOT_AVAILABLE), receive AI ATS match feedback, generate cover letters, track applications along the recruitment pipeline, and submit one-click job applications.

8. **Alumni / Industry Partner:** Submits company profile onboarding requests for Faculty Management approval. Participates in events as an approved attendee/speaker and acts as a recruiter in Vacancy Portal: generating AI job descriptions from bullet points, posting vacancies (subject to Faculty Management approval), managing candidate pipelines, viewing applicant academic GPA/exam records, using natural-language AI candidate search (strictly gated to students with ACTIVELY_LOOKING status), and receiving automated AI candidate recommendations.

Standard HTTP Status Codes & RFC 7807 Error Response

All microservice REST APIs strictly conform to standard HTTP status codes:

Code	Status	Category & Usage Standard
200	OK	Standard response for successful GET, PUT, PATCH, or POST actions returning payloads.
201	Created	Successful entity creation (POST). Includes created entity resource payload.
202	Accepted	Request accepted for asynchronous processing (e.g., batch certificate generation, partner registration request).
204	No Content	Request processed successfully with no payload return (e.g., DELETE operations).
400	Bad Request	Structural validation failure, missing required fields, or malformed syntax.
401	Unauthorized	Missing, expired, or cryptographically invalid JWT authentication token.
403	Forbidden	Authenticated user lacks required Role-Based Access Control (RBAC) permission.
404	Not Found	Requested entity ID or URI endpoint route does not exist.
409	Conflict	Business rule state conflict (e.g., double event booking, duplicate company registration).
422	Unprocessable Entity	Syntactically valid JSON failing domain business validation rules (e.g., end date before start date).
429	Too Many Requests	Rate limit or request quota threshold breached at API Gateway.
500	Internal Error	Unexpected runtime exception. Internal trace details masked for security.
502	Bad Gateway	Gateway received invalid response from downstream microservice.
503	Service Unavailable	Downstream microservice is down or circuit breaker is in OPEN state.

RFC 7807 Standardized Error Payload Structure

```
{
  "type": "[https://portal.domain.com/errors/resource-not-found](https://portal.domain.com/errors/resource-not-found)",
  "title": "Resource Not Found",
  "status": 404,
  "detail": "Event with ID 1024 was not found.",
  "instance": "/api/v1/events/1024",
  "timestamp": "2026-08-18T11:20:00Z"
}
```

Complete REST API Endpoints Specification

1. Auth Service (auth-service) — Port 8081

Controller: AuthController | Base Path: /api/v1/auth

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/auth/staff/invite	Send invitation link to a new staff member	ROLE_ADMIN / ROLE_MANAGEMENT	201, 400, 403, 409
POST	/api/v1/auth/staff/complete-registration	Complete staff account setup via invite token	Public / Invite token	200, 400, 422
POST	/api/v1/auth/partner/pending	Register a new pending industry partner registration request	Public	202, 400, 409
POST	/api/v1/auth/partner/complete-registration	Complete approved partner account registration	Public / Token	200, 400, 422
POST	/api/v1/auth/login	Authenticate Student or Industry Partner credentials	Public	200, 400, 401, 422
POST	/api/v1/auth/staff/login	Staff login Step 1: Triggers 2FA / OTP code dispatch	Public / Staff	200, 401
POST	/api/v1/auth/staff/verify-otp	Staff login Step 2: Verify OTP code & issue stateless JWT	Public / OTP Token	200, 400, 401
POST	/api/v1/auth/validate	Validate JWT token integrity & claims	Query Param / Bearer Header	200, 401
GET	/api/v1/auth/me	Fetch authenticated user context and profile summary	Bearer Token	200, 401
POST	/api/v1/auth/students/sync	Batch auto-provision student accounts from university enrollment database	Internal API / ROLE_ADMIN	200, 401, 403

2. Event Management Service (event-management-service) — Port 8082

Controllers: EventController, VenueController, GuestSpeakerController, AgendaController, CertificateEligibilityCriteriaController

Events (/api/v1/events)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/events	Create a new workshop or industry event	ROLE_COORDINATOR	201, 400, 422
GET	/api/v1/events	List all events (Filterable by ?status=, ?venueId=, ?coordinatorUserId=)	Public / Authenticated	200
GET	/api/v1/events/{id}	Get complete event details by ID	Public / Authenticated	200, 404
PUT	/api/v1/events/{id}	Update event details and schedule	ROLE_COORDINATOR,	200, 400, 404
PATCH	/api/v1/events/{id}/status	Update event lifecycle status (DRAFT, SCHEDULED, COMPLETED, CANCELLED)	ROLE_COORDINATOR	200, 400, 404
PATCH	/api/v1/events/{id}/reschedule	Reschedule event date and time	ROLE_COORDINATOR	200, 400, 404
PATCH	/api/v1/events/{id}/cancel	Cancel a scheduled event	ROLE_COORDINATOR	200, 400, 404
DELETE	/api/v1/events/{id}	Delete an event record	ROLE_COORDINATOR	204, 404
POST	/api/v1/events/{id}/coordinator	Assign a coordinator to an event	ROLE_MANAGEMENT	200, 400, 404
DELETE	/api/v1/events/{id}/coordinator	Remove coordinator from an event	ROLE_MANAGEMENT	204, 404

Venues (/api/v1/venues)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/venues	Create a new campus venue	ROLE_ADMIN_STAFF	201, 400, 409
GET	/api/v1/venues	List all campus venues	Public / Authenticated	200
GET	/api/v1/venues/{id}	Get venue details by ID	Public / Authenticated	200, 404
PUT	/api/v1/venues/{id}	Update venue capacity and details	ROLE_ADMIN_STAFF	200, 400, 404
DELETE	/api/v1/venues/{id}	Delete a venue record	ROLE_ADMIN_STAFF	204, 404

Guest Speakers (/api/v1/guest-speakers)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/guest-speakers	Create a guest speaker profile	ROLE_COORDINATOR	201, 400, 409
GET	/api/v1/guest-speakers	List all guest speakers (Filterable by?organizationId=)	ROLE_COORDINATOR	200
GET	/api/v1/guest-speakers/{id}	Get guest speaker details by ID	Authenticated	200, 404
PUT	/api/v1/guest-speakers/{id}	Update guest speaker profile	ROLE_COORDINATOR, ROLE_SPEAKER	200, 400, 404
DELETE	/api/v1/guest-speakers/{id}	Delete guest speaker profile	ROLE_COORDINATOR	204, 404

Agendas (/api/v1/agendas)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/agendas	Create an agenda session item	ROLE_COORDINATOR	201, 400
GET	/api/v1/agendas	List agenda items (Filterable by ?eventId= or ?speakerId=)	Public / Authenticated	200
GET	/api/v1/agendas/{id}	Get agenda item details by ID	Public / Authenticated	200, 404
PUT	/api/v1/agendas/{id}	Update agenda item details	ROLE_COORDINATOR	200, 400, 404
DELETE	/api/v1/agendas/{id}	Delete agenda item	ROLE_COORDINATOR	204, 404

Certificate Eligibility Criteria (/api/v1/events/{eventId}/eligibility-criteria)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
PUT	/api/v1/events/{eventId}/eligibility-criteria	Create or update certificate eligibility rules for an event	ROLE_COORDINATOR	200, 400, 404
GET	/api/v1/events/{eventId}/eligibility-criteria	Get certificate eligibility rules configured for an event	ROLE_COORDINATOR, ROLE_STAFF	200, 404
DELETE	/api/v1/events/{eventId}/eligibility-criteria	Delete certificate criteria for an event	ROLE_COORDINATOR	204, 404

3. Event Participation Service (event-participation-service) — Port 8083

Controllers: RegistrationController, QrSessionController

Registrations (/api/v1/registrations)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/registrations	Register a participant for an event	ROLE_STUDENT, ROLE_ACADEMIC_STAFF	201, 400, 409, 422
GET	/api/v1/registrations	List all event registrations (Filterable by ?eventId=)	ROLE_COORDINATOR, ROLE_STAFF	200
GET	/api/v1/registrations/{registrationId}	Get registration details by ID	Authenticated	200, 404
PATCH	/api/v1/registrations/{registrationId}/statuses	Update registration status (CONFIRMED, CANCELLED, CHECKED_IN, NO_SHOW)	ROLE_COORDINATOR, ROLE_ADMIN_STAFF	200, 400, 404
DELETE	/api/v1/registrations/{registrationId}	Cancel or delete an event registration	Authenticated	204, 404

QR Sessions & Attendance Verification (/api/v1/...)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/events/{eventId}/qr-sessions	Generate dynamic QR check-in session for an event	ROLE_COORDINATOR	201, 403, 404
GET	/api/v1/events/{eventId}/qr-sessions	List all QR check-in sessions for an event	ROLE_COORDINATOR, ROLE_STAFF	200, 404
GET	/api/v1/qr-sessions/{qrId}	Get QR session details by ID	ROLE_COORDINATOR, ROLE_STAFF	200, 404
GET	/api/v1/qr-sessions/{qrCodeValue}/verify	Verify QR code scanned by an attendee (supports manual override by Admin Staff)	ROLE_STUDENT, ROLE_ADMIN_STAFF, ROLE_COORDINATOR	200, 400, 409, 422
DELETE	/api/v1/qr-sessions/{qrId}	Deactivate a live QR check-in session	ROLE_COORDINATOR	204, 404

4. Application Service (application-service) — Port 8084

Controller: JobApplicationController | Base Path: /api/v1/applications

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/applications	Submit a new job application	ROLE_STUDENT	201, 400, 409, 422
GET	/api/v1/applications/{id}	Get job application details by ID	Authenticated	200, 404
GET	/api/v1/applications/vacancy/{vacancyId}	List all applications for a specific vacancy	ROLE_PARTNER	200, 403, 404
GET	/api/v1/applications/alumni/{alumniId}	List all applications submitted by a student	ROLE_PARTNER	200, 404
PUT	/api/v1/applications/{id}/status	Update application status (APPLIED, SHORTLISTED, SELECTED, REJECTED, PLACED)	ROLE_PARTNER	200, 400, 404
GET	/api/v1/applications/{id}/audits	View application status change audit history	ROLE_PARTNER	200, 404
POST	/api/v1/applications/{id}/stages	Schedule a recruitment pipeline stage (e.g., Interview, Technical Test)	ROLE_PARTNER	201, 400, 404
PUT	/api/v1/applications/{id}/stages/{stageId}	Update recruitment stage status and notes	ROLE_PARTNER	200, 400, 404
GET	/api/v1/applications/{id}/stages	List all recruitment stages for an application	Authenticated	200, 404

5. Notification Service (notification-service) — Port 8088

Controller: NotificationTemplateController | Base Path: /api/v1/templates

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/templates	Create a new HTML notification template	Authenticated	201, 400
GET	/api/v1/templates	List all notification templates	Authenticated	200
GET	/api/v1/templates/{id}	Get notification template by ID	Authenticated	200, 404
GET	/api/v1/templates/code/{templateCode}	Get notification template by unique template code	Authenticated, Internal API	200, 404
PUT	/api/v1/templates/{id}	Update notification template content	Authenticated	200, 400, 404
DELETE	/api/v1/templates/{id}	Delete notification template	Authenticated	204, 404
Async	RabbitMQ Listeners	Asynchronous delivery listeners for notification.otp, notification.invitation, notification.reminder, notification.announcement, notification.update, notification.certificate, notification.job_alert, notification.onboarding	System Listener	N/A

6. Audit & Storage Service (audit-storage-service) — Port 8089

Controllers: StorageController, AuditController

File Storage (/api/v1/storage)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/storage/upload	Upload file (multipart/form-data with file, uploaderId, fileType)	Authenticated	201, 400, 413, 422
GET	/api/v1/storage/download/{id}	Download stored file by ID	Authenticated	200, 404
GET	/api/v1/storage/{id}	Get stored file metadata by ID	Authenticated	200, 404
GET	/api/v1/storage?uploaderId={uploaderId}	List files uploaded by a specific user	Authenticated	200, 404
DELETE	/api/v1/storage/{id}	Delete file from storage	Authenticated	204, 404

Audit Trail (/api/v1/audit)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/audit/log	Ingest and asynchronously record an audit log event	Internal API / ROLE_ADMIN	201, 400
GET	/api/v1/audit/logs	Query paginated immutable audit logs (?userId=, ?action=, ?from=, ?to=)	ROLE_ADMIN, ROLE_MANAGEMENT	200, 403

7. Platform Management Service (platform-management-service) — Port 8086

Partner Verification (/api/v1/admin/partner-verifications)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
GET	/api/v1/admin/partner-verifications?status={status}	List partner verifications by status (Paginated)	ROLE_MANAGEMENT	200, 403
GET	/api/v1/admin/partner-verifications/{id}	Get partner verification record by ID	ROLE_MANAGEMENT	200, 404
GET	/api/v1/admin/partner-verifications/{id}/history	Get decision/approval history for partner verification	ROLE_MANAGEMENT	200, 404
POST	/api/v1/admin/partner-verifications/{id}/claim?reviewerId={id}	Management claims a verification task	ROLE_MANAGEMENT	200, 400, 404
POST	/api/v1/admin/partner-verifications/{id}/decision	Record review decision (APPROVED, REJECTED, CHANGES_REQUESTED)	ROLE_MANAGEMENT	200, 400, 404
PATCH	/api/v1/admin/partner-verifications/{id}	Admin edit partner verification metadata	ROLE_MANAGEMENT	200, 400, 404

Partner Documents & Review Submission (/api/v1/partner-verifications/{verificationId})

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/partner-verifications/{verificationId}/documents	Register/attach a verification document	ROLE_PARTNER	201, 400, 422
GET	/api/v1/partner-verifications/{verificationId}/documents	List attached verification documents	ROLE_PARTNER, ROLE_MANAGEMENT	200, 404
DELETE	/api/v1/partner-verifications/{verificationId}/documents/{documentId}	Remove an attached document	ROLE_PARTNER	204, 404
POST	/api/v1/partner-verifications/{verificationId}/submit-for-review	Submit document package for review	ROLE_PARTNER	200, 400, 422

Partner Verification (Internal Service-to-Service)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/internal/partner-verifications	Internal hook to create governance record on new partner signup	Internal API	201, 400

Vacancy Approvals (/api/v1/admin/vacancy-approvals)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
GET	/api/v1/admin/vacancy-approvals?status={status}	List vacancy approvals by status (Paginated)	ROLE_MANAGEMENT	200, 403
GET	/api/v1/admin/vacancy-approvals/{id}	Get vacancy approval record by ID	ROLE_MANAGEMENT	200, 404
GET	/api/v1/admin/vacancy-approvals/{id}/history	Get vacancy approval decision history	ROLE_MANAGEMENT	200, 404
POST	/api/v1/admin/Vacancy-approvals/{id}/claim?reviewerId={id}	Admin or Management claims a vacancy review task	ROLE_MANAGEMENT	200, 400, 404
POST	/api/v1/admin/vacancy-approvals/{id}/decision	Record vacancy decision (APPROVED, REJECTED, CHANGES_REQUESTED)	ROLE_MANAGEMENT	200, 400, 404
PATCH	/api/v1/admin/vacancy-approvals/{id}	Admin edit vacancy submission data	ROLE_MANAGEMENT	200, 400, 404

Vacancy Approvals (Internal Service-to-Service)

HTTP Method	Endpoint Path	Description	Access / RBAC Role	Expected Status Codes
POST	/api/v1/internal/vacancy-approvals	Internal hook to submit vacancy for governance check	Internal API	201, 400

8. User Service (user-service) — Port 8087

HTTP Method	Endpoint Path	Target Description	Planned RBAC Role
GET	/api/v1/users/{id}/profile	Retrieve user profile metadata	Authenticated
PUT	/api/v1/users/{id}/profile	Update basic user profile & bio	Authenticated
POST	/api/v1/users/{id}/resumes	Attach uploaded resume metadata	ROLE_STUDENT
PATCH	/api/v1/users/{id}/job-status	Toggle job availability (ACTIVELY_LOOKING / NOT_AVAILABLE)	ROLE_STUDENT
GET	/api/v1/students/{studentId}/academic-record	Fetch student GPA & exam results	ROLE_STUDENT, ROLE_PARTNER

9. Vacancy Service (vacancy-service) — Port 8090

HTTP Method	Endpoint Path	Target Description	Planned RBAC Role
GET	/api/v1/vacancies	Multi-criteria vacancy search	Public / Authenticated
POST	/api/v1/vacancies	Corporate partner submits new job vacancy	Authenticated
PUT	/api/v1/vacancies/{id}	Update job posting details	Authenticated
DELETE	/api/v1/vacancies/{id}	Deactivate or archive job vacancy	Authenticated

C. Certificate Service (certificate-service) — Port 8085

HTTP Method	Endpoint Path	Target Description	Planned RBAC Role
POST	/api/v1/certificates/templates	Upload PDF certificate background template	Authenticated
POST	/api/v1/certificates/generate	Batch generate PDF certificates for eligible attendees	Authenticated
GET	/api/v1/certificates/student/{studentId}	Fetch earned certificates for a student	Authenticated
GET	/api/v1/certificates/verify/{qrHash}	Public QR verification lookup for third parties	Public

RabbitMQ Event Bus Topology

1. Exchange & Queue Architectural Design

- **Main Exchange:** notification.exchange (Type: Topic Exchange, Durable: true)
- **Dead Letter Exchange (DLX):** notification.dlx (Type: Direct Exchange, Durable: true)

2. Complete Notification Service Queue & Routing Matrix

Queue Name	Routing Key	Dead Letter Queue	Target Event Description
otp.queue	notification.otp	otp.dlq	Consumes 2FA OTP codes for staff login dispatch.
invitation.queue	notification.invitation	invitation.dlq	Consumes single-use email invites for guest speakers and staff.
reminders.queue	notification.reminder	reminders.dlq	Consumes pre-session reminders (2 days prior to events).
announcement.queue	notification.announcement	announcement.dlq	Consumes mass event broadcasts & maintenance notices.
update.queue	notification.update	update.dlq	Consumes candidate application status change alerts.
certificate.queue	notification.certificate	certificate.dlq	Consumes issued PDF certificate download links.
job_alert.queue	notification.job_alert	job_alert.dlq	Consumes AI candidate sourcing & job posting alerts.
onboarding.queue	notification.onboarding	onboarding.dlq	Consumes partner onboarding approval/rejection decision emails.

Additional Inter-Service Domain Events

Topic Key	Publisher Service	Consumer Service(s)	Business Purpose
event.completed	Event Management Service	Certificate Service	Automatically triggers PDF certificate generation post-event.
partner.approved	Platform Management Service	Auth Service, Notification Service	Activates partner credentials and dispatches welcome email.
vacancy.created	Vacancy Service	Platform Management Service	Submits newly created job vacancy to admin review queue.
vacancy.approved	Platform Management Service	Vacancy Service, Notification Service	Publishes approved vacancy live to student search index.
ai.sourcing	AI Service	Notification Service	Alerts high-match candidate pool of suitable job openings.

Status / Feedback Queues (Return Path to Producer Services)

Status Queue Name	Status Routing Key	Purpose
otp.status.queue	notification.status.otp	Emits email delivery status log back to Auth Service.
invitation.status.queue	notification.status.invitation	Emits invite dispatch verification back to Event Service.
reminder.status.queue	notification.status.reminder	Emits reminder delivery log back to Event Service.
announcement.status.queue	notification.status.announcement	Emits broadcast completion log to Platform Mgmt Service.
update.status.queue	notification.status.update	Emits application update delivery log to Application Service.